

ST Bus Ticket Booking System — Comprehensive Lab Testing Report

1. High-Level Test Case Index

Pre-condition: ST Web/Mobile application should be accessible and server infrastructure is active.

Test ID	Test Title	Test Type	Description	Priority
TC_ST_001	Search bus with valid route & date	Functional	Verify system returns available ST buses when valid locations are entered	High
TC_ST_002	Search bus with invalid route criteria	Functional	Verify system shows validation errors for empty fields or matching locations	High
TC_ST_003	Search bus when no schedules exist	Functional	Verify system displays "No buses available for this route" cleanly	Medium
TC_ST_004	Verify UI elements on Search Bus page	GUI	Verify all dropdowns, date pickers, and buttons display correctly	Low

2. Detailed Test Cases Execution Logs

Test Case Matrix: TC_ST_001

Test Case ID	TC_ST_001
Test Priority	High
Test Case Title	Search bus with valid route and travel date
Test Type	Functional
Pre-condition	Passenger application is open and 'Search Bus' page is fully loaded.

Step	Description	Expected Result	Actual Result	Status	Data	Bug ID
1	Select valid 'From' station dropdown.	Source station field should accept input.	Input accepted successfully.	Pass	From: 'Mumbai Central'	-
2	Select valid 'To' station dropdown.	Destination station field should accept input.	Input accepted successfully.	Pass	To: 'Pune Swargate'	-
3	Choose a valid upcoming journey date.	Date picker should capture selection.	Selected date locked in calendar.	Pass	Date: Future Date	-

4	Click on the 'Search Buses' button.	System should query and display matching ST bus trips.	Available buses displayed successfully.	Pass	-	-
5	Verify bus card details.	Bus cards show Departure Time, Type (Shivneri, Ordinary), Fare, and Seats.	List view correctly displays schedule parameters.	Pass	List of ST Buses	-

Test Case Matrix: TC_ST_002

Test Case ID	TC_ST_002
Test Priority	High
Test Case Title	Search bus with invalid route criteria
Test Type	Functional (Negative)
Pre-condition	Passenger application is open and 'Search Bus' page is fully loaded.

Step	Description	Expected Result	Actual Result	Status	Data	Bug ID
1	Select identical stations for 'From' and 'To' fields.	System should flag an immediate error validation.	Page allowed identical route selection without inline warning.	Fail	-	BUG_ST_101
2	Leave the journey date field empty and click 'Search'.	System should trigger standard validation message.	'Please choose a journey date' warning appeared.	Pass	Date: [Blank]	-
3	Type validation symbols inside station fields.	Field must scrub text input or display zero results.	Dropdown empty; no arbitrary text saved.	Pass	Input: '@@@@123'	-

Test Case Matrix: TC_ST_003

Test Case ID	TC_ST_003
Test Priority	Medium
Test Case Title	Search bus when no schedules exist
Test Type	Functional (Edge Case)
Pre-condition	System database contains zero planned ST schedules for the queried route/day.

Step	Description	Expected Result	Actual Result	Status	Data	Bug ID
1	Select valid low-traffic 'From' station.	Input accepted successfully.	Input accepted successfully.	Pass	From: 'Gaganbawda'	-
2	Select valid low-traffic 'To' station.	Input accepted successfully.	Input accepted successfully.	Pass	To: 'Toranmal'	-

3	Enter valid journey date.	Input accepted successfully.	Input accepted successfully.	Pass	Date: Mid-week date	-
4	Click on 'Search Buses' button.	System handles absence of data gracefully.	Screen accurately loaded empty-state message.	Pass	Message: 'No ST buses found'	-

Test Case Matrix: TC_ST_004

Test Case ID	TC_ST_004
Test Priority	Low
Test Case Title	Verify UI elements of the Search Ride page
Test Type	GUI / Usability
Pre-condition	Home landing screen / dashboard is successfully requested.

Step	Description	Expected Result	Actual Result	Status	Data	Bug ID
1	Check 'From' and 'To' dropdown visual boundaries.	Fields should be aligned, visible, and clearly legible.	Fields visible and scaled accurately.	Pass	UI Layout Validated	-
2	Inspect Calendar / Date-Picker object icon.	Calendar button should be present inside/near field.	Icon rendered properly inside elements.	Pass	Icon Displayed	-
3	Validate 'Search Buses' CTA button layout.	Button text is centered and clearly clickable.	Button display matches mockups perfectly.	Pass	Hover state active	-
4	Check 'Check PNR / Track Bus' widget availability.	Live tracker or PNR sub-elements are visible.	Layout sections load smoothly.	Pass	Widget Visible	-